CROSCIM · multiscale, observation-conditioned generative modelling of sea ice

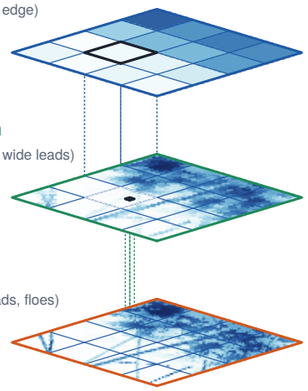
Nested levels $\mathcal{C}$ (32 km) $\supset$ $\mathcal{M}$ (4 km) $\supset$ $\mathcal{H}$ (1 km) on a polar-stereographic grid · native L2 observations → level latents → cascade of conditional flow-matching generators → analysis, rollout and scale-interaction experiments

A Nested grids, fixed operators, scale-specific state

$\mathcal{C}$ · Δ = 32 km
$X_{\mathcal{C}}$: 4×4 cells (ice edge)

$\mathcal{M}$ · Δ = 4 km
$X_{\mathcal{M}}$: 32×32 (MIZ, wide leads)

$\mathcal{H}$ · Δ = 1 km
$X_{\mathcal{H}}$: 128×128 (leads, floes)



same 128-km window; black: one $\mathcal{C}$ cell → its 64 $\mathcal{M}$ children; one $\mathcal{M}$ cell → its 16 $\mathcal{H}$ children

Restriction / prolongation (fixed, not learned)

$$(Rx)(k) = \frac{\sum_{i \in \mathrm{ch}(k)} \omega_i x(i)}{\sum_{i \in \mathrm{ch}(k)} \omega_i}$$

$$(Px)(i) = x(\mathrm{par}(i)), \quad RP = I$$

ω_i: ocean fraction; equal-area cells → conservative

Scale-specific degrees of freedom

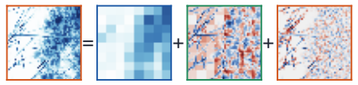
$Y = (X_{\mathcal{C}}, \delta X_{\mathcal{M}}, \delta X_{\mathcal{H}})$

$\delta X_{\mathcal{M}} = X_{\mathcal{M}} - PX_{\mathcal{C}}, \quad \delta X_{\mathcal{H}} = X_{\mathcal{H}} - PX_{\mathcal{M}}$

$R\delta X = 0$ (non-redundant)

$X_{\mathcal{H}} = P_{\mathcal{C}\to\mathcal{H}} X_{\mathcal{C}} + P_{\mathcal{M}\to\mathcal{H}} \delta X_{\mathcal{M}} + \delta X_{\mathcal{H}}$

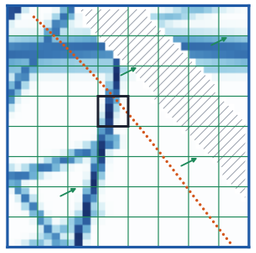
exact recomposition (256-km region, synthetic SIC):



B Observations → level latents: tokenization and attention

one $\mathcal{C}$ cell k (32 km), window T = 3 d

SAR SIC 1 km (hatched: gap), CRISTAL track, SAR drift; green: 64 $\mathcal{M}$ cells



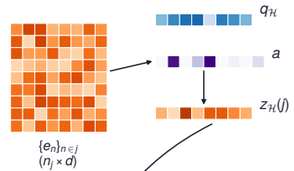
$N_{\mathrm{obs}} \sim 10^3$–$10^4$ per $\mathcal{C}$ cell

$\ell(n) = \arg\min_{\ell} |\log r_n - \log \Delta_\ell|$

cost $O(N_{\mathrm{obs}} d) + O(64Kd)$, once per sample

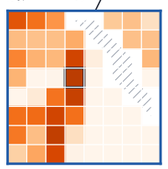
Step 1 · set encoder in $\mathcal{M}$ cell j

$e_n = \varphi(v_n, \sigma_n, \Delta\mathbf{x}_n, \Delta t_n, s_n) \in \mathbb{R}^d$



$a = \mathrm{softmax}(q_{\mathcal{H}} W_K E^{\top} / d^{1/2})$

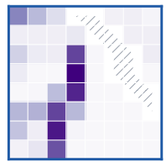
$z_{\mathcal{H}}(j) = a\, E\, W_V$; dense: masked patch embed



$z_{\mathcal{H}}$ on $\mathcal{M}$ grid: 64 tokens per $\mathcal{C}$ cell, any N_{obs}

Step 2 · $\Pi_\uparrow$ to cell k

α_{kj} (illustrative)



$q_k = q_0 + W_O z_{\mathcal{C}}(k)$

$\alpha_{kj} = \mathrm{softmax}_j(q_k^{\top} W_K \tilde{z}_j / d^{1/2})$

$\tilde{z}_j = z_{\mathcal{H}}(j) + p_j$ (child position)

$(\Pi_\uparrow z_{\mathcal{H}})(k) = \Sigma_j \alpha_{kj} W_V \tilde{z}_j$

→ $c_{\mathcal{C}}(k)$
K = 2 vectors $\in \mathbb{R}^d$
+ coverage, count, mean, var

C Conditioning groups and chain rule = posterior

	obs $\mathcal{C}$	obs $\mathcal{M}$	obs $\mathcal{H}$	coarser	own past	forcing
$c_{\mathcal{C}}$	$z_{\mathcal{C}}$	$\Pi_\uparrow z_{\mathcal{M}}$	$\Pi_\uparrow z_{\mathcal{H}}$	—	$s^t_{\mathcal{C}}, \Pi_\uparrow s^t_{\mathcal{M},\mathcal{H}}$	F
$c_{\mathcal{M}}$	$\Pi_\downarrow z_{\mathcal{C}}$	$z_{\mathcal{M}}$	$\Pi_\uparrow z_{\mathcal{H}}$	$X_{\mathcal{C}}$	$s^t_{\mathcal{M}}, \Pi_\uparrow s^t_{\mathcal{H}}$	$\Pi_\downarrow F$
$c_{\mathcal{H}}$	$\Pi_\downarrow z_{\mathcal{C}}$	$\Pi_\downarrow z_{\mathcal{M}}$	$z_{\mathcal{H}}$	$X_{\mathcal{C}}, \delta X_{\mathcal{M}}$	$s^t_{\mathcal{H}}$	$\Pi_\downarrow F$

filled: own level · tinted: shared · black frame: small → large (bottom-up) · own past: Stage 2 only

groups are concatenated, never summed → each can be dropped (∅, group dropout) or replaced

(a) exact in observation space:

$p(Y|O) = p(X_{\mathcal{C}}|O)\, p(\delta X_{\mathcal{M}}|X_{\mathcal{C}}, O)\, p(\delta X_{\mathcal{H}}|X_{\mathcal{C}}, \delta X_{\mathcal{M}}, O)$

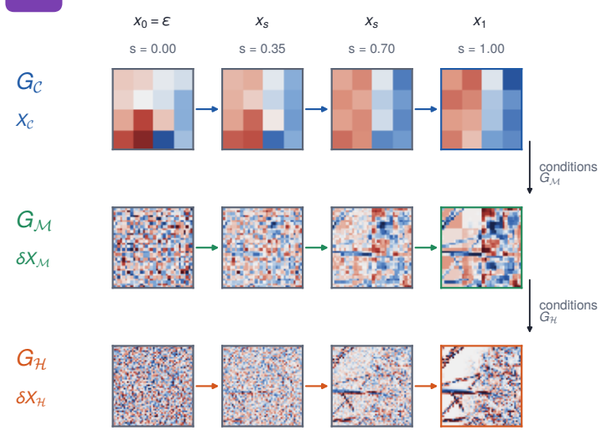
(b) model, with summaries $c_\ell = g_\ell(O)$:

$p_\theta(Y|O) = p_\theta(X_{\mathcal{C}}|c_{\mathcal{C}})\, p_\theta(\delta X_{\mathcal{M}}|X_{\mathcal{C}}, c_{\mathcal{M}})\, p_\theta(\delta X_{\mathcal{H}}|X_{\mathcal{C}}, \delta X_{\mathcal{M}}, c_{\mathcal{H}})$

(c) exact iff $Y_\ell \perp O \,|\, (Y_{<\ell}, g_\ell(O))$ (sufficiency; needs $\Pi_\uparrow$ in $c_{\mathcal{C}}$)

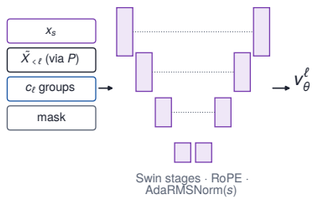
(d) FM: $\mathcal{L}^*(g) - \mathcal{L}^*(O) = \mathbb{E}\|v^*_O - v^*_g\|^2 \geq 0$, $= 0$ iff same conditional law

D Conditional flow matching, one generator per level (cascade)



normalized units; noise for δ levels projected ($R\varepsilon = 0$); illustrative x_s from true x_1

Velocity net v^ℓ_θ (Swin U-Net)



$x^\ell_s = s\, x^\ell_1 + (1-s)\,\varepsilon^\ell, \quad u = x^\ell_1 - \varepsilon^\ell$

$\mathcal{L}_\ell = \mathbb{E}[m_\ell \odot (\frac{1}{2}\| \frac{u - v^\ell_\theta}{\sigma_\ell} \|^2 + \log \sigma_\ell)]$

- $\varepsilon, v \leftarrow (I - PR)\cdot$ for δ levels
- $\tilde{X}_{<\ell} = X_{<\ell} + \eta$ (truth → samples)
- m_ℓ: observed cells; σ_ℓ per level
- bounds: censoring, recomposed field
- Heun sampler, one member at a time
- no learned decoder (P recomposes)

E Dataloader and training

batch geometry



$\mathcal{C}$: full domain
$\mathcal{M}$: window + halo (green)
$\mathcal{H}$: window in ROI (orange)

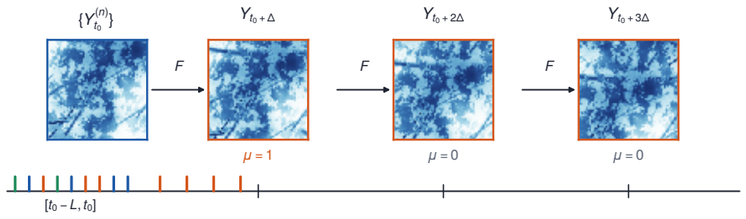
per sample (t_0, window)

1. obs tables (Zarr, one per stream) + time index: select $[t_0 - L, t_0]$ and $(t, t + \Delta]$, no interpolation
2. split input / target by stream, region or time
3. inputs → tokens per level (patches + masks; packed ragged arrays for irregular obs)
4. targets → binned per level (R on obs) + masks m_ℓ
5. ERA5 forcing dense on $\mathcal{C}$; $\Pi_\uparrow z_{\mathcal{H}}$ outside the $\mathcal{H}$ window precomputed or no-grad

training

1. Stage 1 on observation windows (levels in parallel, teacher forcing + η)
2. archive of analysis ensembles
3. Stage 2 on those analyses: pushforward 2–4 steps, memory σ_ℓ
- group dropout: obs groups (μ), $\Pi_\uparrow$ groups → ∅
- held-out streams kept for verification only

F Analysis, rollout and re-conditioning



Stage 1: $p(Y_{t_0}|z(O_{t_0-L:t_0}), F)$ · Stage 2: $p(Y^{t+\Delta}|Y^t, F_{t:t+\Delta}, [z(O_{(t,t+\Delta]})])$

$\mu = 1$: re-conditioning by new obs (learned filter) · $\mu = 0$: forecast; state physical at every step

Small scales without observations

memory: $x^\ell_0 = Y^t_\ell + \sigma_\ell \varepsilon$ (small $\sigma_{\mathcal{C}}$: persistence; large $\sigma_{\mathcal{H}}$: refresh)

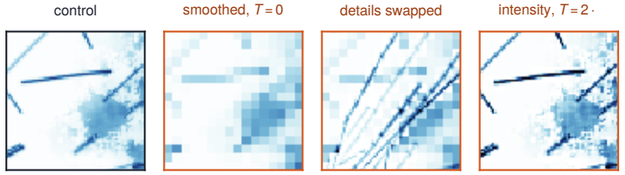
$t < \tau_\ell$: prognostic (carried, advected) $t > \tau_\ell$: $\delta X_{\mathcal{H}} \sim p(\delta X_{\mathcal{H}}|X_{\mathcal{C}}, \delta X_{\mathcal{M}}, F)$

two modes: (i) ROI, short range: $\mathcal{H}$ every step, closure $\Pi_\uparrow s^t_{\mathcal{H}}$;
(ii) long range / outside ROI: $\mathcal{C}$ carries sub-grid moments $\phi_{\mathcal{C}}$ (learned ITD-like state)

G Scale-interaction experiments: interventions that preserve R, paired noise

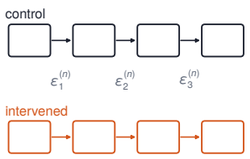
$\delta X'_{\mathcal{H}} = \Pi_0\, T(\delta X_{\mathcal{H}}), \quad \Pi_0 \delta = \delta - PR\delta \Rightarrow X_{\mathcal{M}}, X_{\mathcal{C}}$ unchanged at t

applied to Y^t (dynamical path) or to $O_{\mathcal{H}}$ (observational path), then re-encoded



all four panels: same $X_{\mathcal{M}}$ and $X_{\mathcal{C}}$ (synthetic SIC)

paired rollouts



same noise ⇒ difference = effect of the intervention only

$I_\uparrow(\tau) = \|\overline{X^{int}_{\mathcal{C}} - X^{ctrl}_{\mathcal{C}}}(t+\tau)\| / \sigma_{\mathcal{C}}(t+\tau)$

closure off: $\Pi_\uparrow s^t_{\mathcal{H}}, \Pi_\uparrow z_{\mathcal{H}} \to \varnothing$ in $c_{\mathcal{C}}$

large → small: $X'_{\mathcal{C}} = X_{\mathcal{C}} + \Delta X_{\mathcal{C}}$, regenerate,
$I_\downarrow = \|\mathbb{E}S(\delta X^{int}_{\mathcal{H}}) - \mathbb{E}S(\delta X^{ctrl}_{\mathcal{H}})\| / \sigma_S$

real feedback: $I_\uparrow(\tau)$ persists beyond $\tau_{\mathcal{H}}$

controls: placebo swaps (same statistics), maps, spectra, by season and region

H Data-assimilation reading and diagnostics

Information form: $p(X|y) \propto p(X) \exp(X^{\top} H^{\top} R^{-1} y - \frac{1}{2} X^{\top} H^{\top} R^{-1} H X)$

$(H^{\top} R^{-1} y,\ H^{\top} R^{-1} H)$ is sufficient and lives in state space → the encoder plays $H^{\top} R^{-1}$, not H

Coarse aggregation: $\hat{x}_k = \Sigma_i w_i y_i / \Sigma_i w_i, \quad \mathrm{Var} = (\Sigma_i w_i)^{-1}, \quad w_i = (\sigma_i^2 + \sigma_\delta^2)^{-1}$

$\Pi_\uparrow$ has this form ($\alpha_{ki} \propto w_i$, representativeness included); $K > 1$: higher sub-grid moments

DA	CROSCIM
H: state → obs	explicit only in the loss (m_ℓ, binning R)
$H^{\top} R^{-1} y,\ H^{\top} R^{-1} H$	$z_\ell = E_\ell(y)$; coverage, count, σ
analysis (3D/4D-Var, EnKF)	Stage 1: amortized posterior sampler
forecast + update	Stage 2, fused; $\mu \in \{0, 1\}$
OSE / observation impact	group dropout, one stream at a time

Diagnostics

- prior ($\mu = 0$) vs posterior ($\mu = 1$), increments
- spread reduction, per-stream impact
- Desroziers, rank histograms, $(y - H\hat{X})/\sigma$
- CRPS, spread–skill, PSD per level, τ_ℓ
- explicit-H guidance for new sensors:
  $v \leftarrow v + \lambda_s \nabla \log p(y|H(\hat{x}_1))$ (→ 4DVarNet)